MATHEMATICS

1. $a = \sqrt{3}$

$d = \sqrt{3}$

$a_{100} = a + (100-1)d$

$a_{100} = \sqrt{3} + (100-1)\sqrt{3}$

$a_{100} = \sqrt{3} + 99\sqrt{3}$

$\boxed{a_{100} = 100\sqrt{3}}$

2. A circle can have infinitely many tangents

3. n^{th} term $= 8n - 5$

$a_1 = 8 \times 1 - 5$
$= 3$

$a_2 = 8 \times 2 - 5$
$= 11$

$d = 11 - 3$
$= 8$

$S_{28} = \frac{28}{2}[2a + (28-1)d]$

$14[2 \times 3 + 27 \times 8]$

$14[6 + 216]$

14×222

$S_{28} = \underline{3108}$

4. In Δ POQ

$\angle PQO = 55°$

$\angle POQ = 90°$

$\angle OPQ = 180° - (90° + 55°)$

$= 180 - 145$

$= \cancel{145}\ 35°$

$\angle OPT = 90°$ — (Radius drawn at the point of contact on the tangent is perpendicular

$\angle QPT = 90° - 35°$

$\boxed{= 55°}$

5. $a = 3$

$d = 12$

$a_{54} = a + 53d$

$3 + 53 \times 12$

$= 3 + 636 = 639$

$132 + 639 = 771$

$771 = 3 + (n-1)12$

$768 = (n-1)12$

$64 = n - 1$

$n = 65$

Sol^n. The <u>65th</u> term

6. In ΔORP

$OP = 5\,cm$

$PR = 12\,cm$

Applying Pythagoras

$OR^2 = OP^2 + PR^2$

$OR^2 = (5)^2 + (12)^2$

$OR^2 = 25 + 144$

$OR^2 = \cancel{179}\ 169$

$OR = \sqrt{169}$

$OR = 13\,cm$

In ΔOQR, Applying Pythagoras

$OR = 13$

$OQ = 4\,cm$

$RQ^2 = OR^2 - OQ^2$

$RQ^2 = (13)^2 - (4)^2$

$RQ^2 = 169 - 16$

$RQ^2 = 153$

$RQ = \sqrt{153}$

7. The first ~~to~~ three digit no. divisible by 87 is 174 & the last term is 957.
So, the A.P formed is

$174, 261 \ldots 957$

$a_n = a + (n-1)d$

$957 = 174 + (n-1)87$

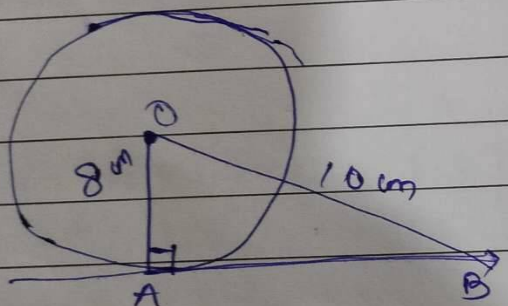
$783 = (n-1)87$

$9 = (n-1)$

$n = 10$

$\therefore$ 10 three digit number are there which is divisible by 87.

8.



In ΔOAB

Applying Pythagoras

$AB^2 = OB^2 - OA^2$

$AB^2 = (10)^2 - (8)^2$

$AB^2 = 100 - 64$

$AB^2 = 36$

$\boxed{AB = 6 \text{ cm}}$

Length of the tangent is 6 cm

9. 2^{nd} term is 14.
3^{rd} term is 18

$a + d = 14$ - (1)
$a + 2d = 18$ - (2)
Subtract (1) from (2).

$$\cancel{a} + 2d = 18$$
$$-\cancel{a} + d = -14$$
$$d = 4 \quad \text{- (3)}$$

Putting (8) into (1)

$$a + 4 = 14$$
$$a = 10$$

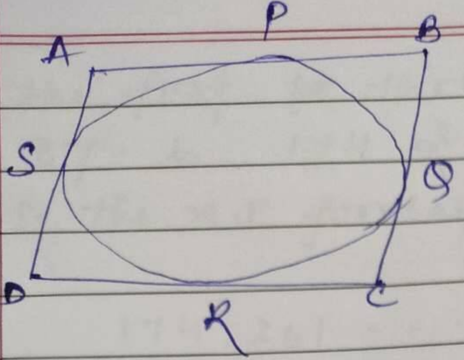
$$S_{51} = \frac{51}{2}[2 \times 10 + 50 \times 4]$$

$$S_{51} = \frac{51}{2}[20 + 200]$$

$$S_{51} = \frac{51}{\cancel{2}} \times \cancel{220}^{110}$$

$$\boxed{S_{51} = 5610}$$

10.



Given : ~~PQ~~ ABCD is a ||gram

To prove : ~~ABCD~~ ABCD is a Rhombus

~~AB = AD~~

~~Adj~~

Proof: Tangents drawn from the same external point are equal

$\therefore$ $AS = AP$ — (1)

$DS = DR$ — (2)

$QC = CR$ — (3)

$QB = PB$ — (4)

~~Add~~ (1), (2), (3) & (4)

$AS + DS + QC + QB = AP + DR + CR + PB$

$AD + BC = AB + DC$

$AD + AD = AB + AB$

$\cancel{2}AD = \cancel{2}AB$

$AD = AB$

(In a ||gram opposite sides are equal)

~~In~~ A ||gram where adjacent sides are equal is Rhombus.

Thus, ABCD is a rhombus